Astrum-1 Hardware Specification Report

Overview

The Astrum-1 system is a 1U CubeSat prototype designed for structural stability, autonomous solar tracking, and high-frequency IoT telemetry. The hardware architecture utilizes a robust power distribution network and high-precision sensing modules to operate reliably in simulated orbital environments.

1. Core Processing & Control

- **Microcontroller (MCU): ESP32-S3-WROOM-1**
  - Features dual-core Xtensa® 32-bit LX7 CPU.
  - Provides integrated 2.4 GHz Wi-Fi and Bluetooth® Low Energy (BLE) for high-speed local data telemetry.
  - Chosen for its advanced power management and dedicated AI acceleration instructions.

2. Telemetry & Communications

- **LoRa Transceiver: AI-Thinker Ra-01SCHP**
  - Based on the Semtech LLCC68 chipset.
  - Operates at 433 MHz with Chirp Spread Spectrum (CSS) modulation.
  - High link budget capacity for long-range, interference-resistant data transmission.

3. Sensors & Data Acquisition

- **Attitude Determination System (ADS): MPU6050**
  - Integrated 6-axis MotionTracking device (3-axis gyroscope + 3-axis accelerometer).
  - Used for calculating real-time pitch and roll orientation for satellite stabilization.
- **Environmental Diagnostics: DHT11**
  - Digital temperature and humidity sensor for monitoring internal structural conditions.
- **Solar Irradiance Sensing: LDR (Light Dependent Resistor)**
  - Configured with 11dB attenuation to track solar vectors and trigger autonomous mechanical solar array adjustments.

4. Actuation & Mechanical Subsystem

- **Servo Motors: SG90 Micro Servos**
    - Dual-servo configuration for independent array articulation.
    - PWM-controlled for precise movement within the 45° to 135° operational range.
    - Power optimized to operate within the 5V logic and structural stress limits.

5. Power & Distribution

- **Logic Voltage:** 3.3V (Logic level) / 5V (Servo/Peripheral power)
- **Energy Source:** Regulated 5V supply line with capacitor filtering to minimize switching noise during high-frequency telemetry bursts.

6. System Interconnects

- **I2C Protocol:** Dedicated SDA (GPIO 21) and SCL (GPIO 20) lines for synchronous sensor data ingestion.
- **PWM Channels:** Independent timers allocated for servo jitter mitigation.